WINDOWS 10/11

Workstation Troubleshooting Project Lab

IT SUPPORT PORTFOLIO PROJECT

COMPLETED

Prepared by	Summer Baham
Completion date	October 6, 2026
Platform	Microsoft Windows 11 Pro
Target role	Service Desk Engineer / IT Support
Completed scope	System inventory, performance diagnostics, startup review, IP configuration, gateway testing, Internet connectivity testing, and DNS verification.

Portfolio integrity note: This completed version documents only the hands-on tasks demonstrated by the attached evidence. Advanced remediation such as DISM/SFC repair, Windows Update repair, driver remediation, and application repair is not claimed as performed because no corresponding fault required those actions in this session.

1. Project Overview

Objective: Perform and document a repeatable Windows workstation diagnostic workflow similar to a service desk assessment. The project demonstrates how to gather system information, review performance and startup behavior, validate IP configuration, test the default gateway and Internet connectivity, verify DNS resolution, and document findings clearly.

Skills Demonstrated

- Windows 11 system information collection using System Information (msinfo32).
- Task Manager performance analysis across CPU, memory, disk, Wi-Fi, and GPU resources.
- Startup application review to identify enabled and disabled programs.
- Command Prompt troubleshooting with ipconfig, ping, and nslookup.
- Layered network isolation: local IP configuration, default gateway, public IP reachability, and DNS resolution.
- Service-desk style documentation, verification, and evidence capture.

Test Environment

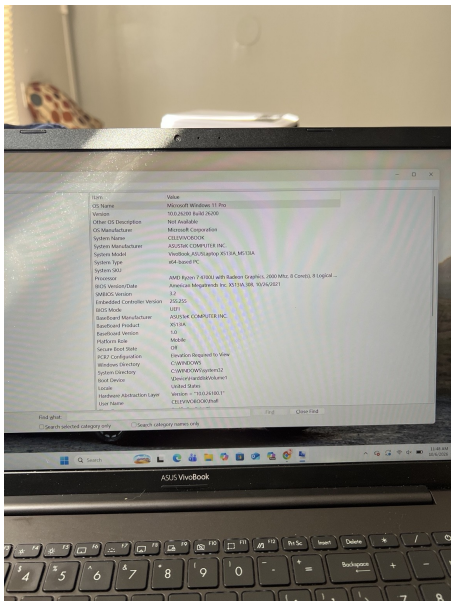
Component	Observed environment
Operating system	Microsoft Windows 11 Pro, Version 10.0.26200, Build 26200
Computer	ASUS VivoBook X513IA / M513IA
Processor	AMD Ryzen 7 4700U with Radeon Graphics, 8 cores / 8 logical processors
Memory	16.0 GB installed RAM (15.4 GB usable shown in Task Manager)
Storage	NVMe SSD, approximately 954 GB formatted capacity shown in Task Manager
Network	Wi-Fi connection; private IPv4 address 192.168.12.134 with gateway 192.168.12.1

Outcome

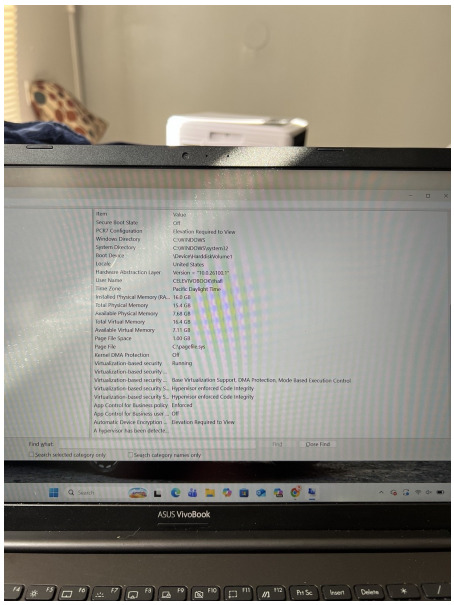
The workstation was successfully inventoried and tested. Performance indicators captured during the session were within normal ranges, the local gateway responded without packet loss, an external IP address was reachable without packet loss, and google.com resolved and responded successfully. DNS lookup also returned valid IPv4 and IPv6 records. No network outage was reproduced during verification.

2. System Information and Baseline Assessment

System Information was opened to establish a baseline before troubleshooting. The evidence confirms Windows 11 Pro, the ASUS VivoBook platform, AMD Ryzen 7 4700U processor, and 16 GB installed physical memory. Establishing the baseline helps a technician confirm the operating system, hardware platform, architecture, and available resources before changing settings.



Evidence A - OS, computer model, processor, BIOS, and system type.



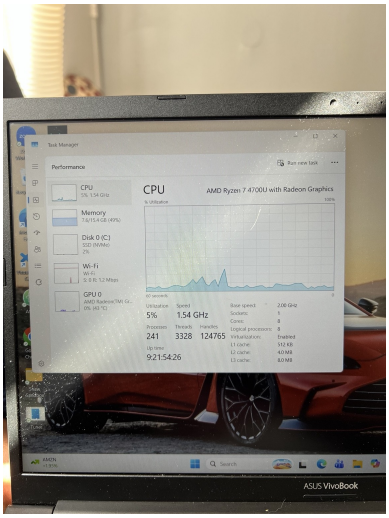
Evidence B - Installed memory and Windows system details.

Baseline Findings

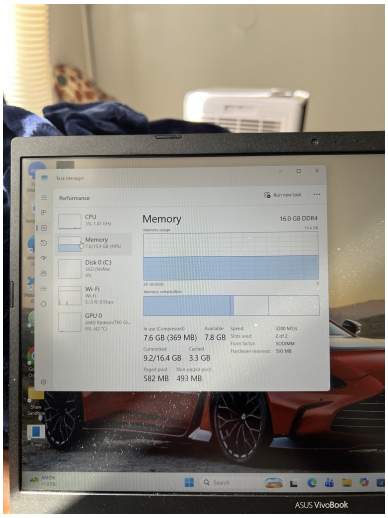
- **Operating system:** Windows 11 Pro, Version 10.0.26200 Build 26200.
- **Processor:** AMD Ryzen 7 4700U with Radeon Graphics.
- **Installed memory:** 16.0 GB RAM.
- **System type:** x64-based PC.
- **Virtualization-based security:** shown as running in the captured System Information view.

3. Performance and Startup Diagnostics

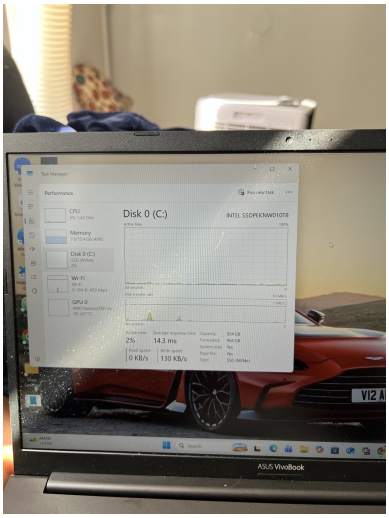
Task Manager was used to review system resource utilization and startup configuration. This is a standard first-line service desk step when a user reports slow startup, lag, or poor application performance.



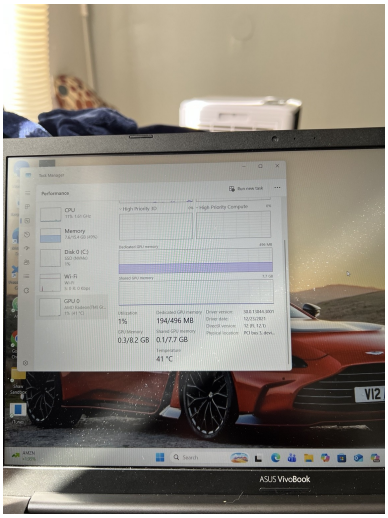
CPU view - low utilization at time of capture; 8 cores and 8 logical processors displayed.



Memory view - approximately 7.6/15.4 GB in use (49%) with about 7.8 GB available.



Disk view - NVMe SSD with low active time during the captured interval.



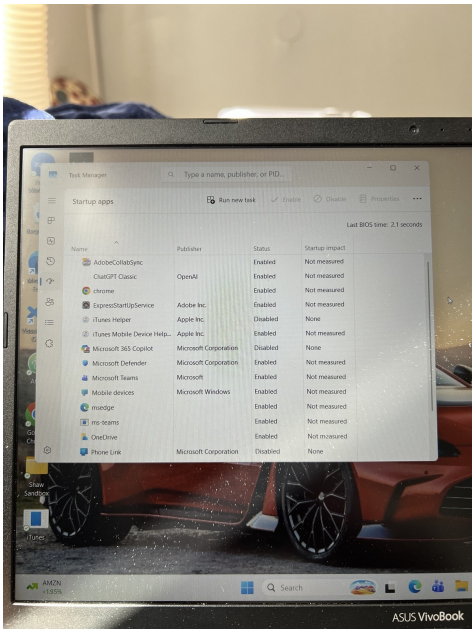
GPU view - low utilization and normal temperature shown at time of capture.

Interpretation

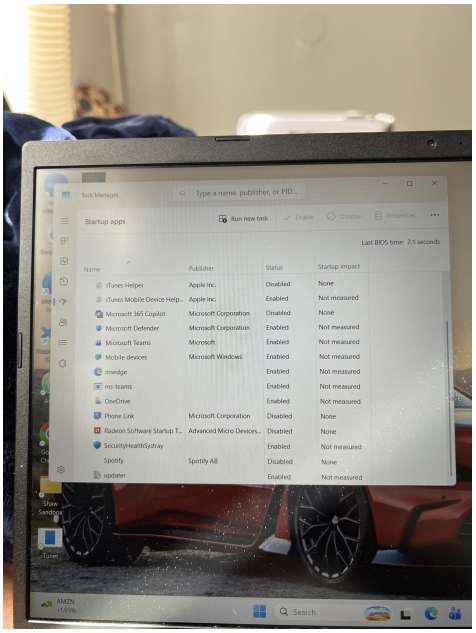
- CPU utilization was low during the capture, so sustained CPU saturation was not observed.
- Memory utilization was approximately 49%, leaving substantial available memory.
- Disk active time was low during the capture, so sustained storage saturation was not observed.
- GPU utilization was minimal during the capture and did not indicate a graphics workload bottleneck.

4. Startup Application Review

The Startup apps section of Task Manager was reviewed to identify applications configured to launch automatically. The purpose of this step is to distinguish required security/system components from optional user applications that may increase sign-in load. No program was disabled solely for the sake of the lab.



Startup apps - upper portion of the list, showing a mix of enabled and disabled entries.



Startup apps - lower portion of the list, continuing the enabled/disabled review.

Service Desk Judgment

A technician should verify business need and user impact before disabling startup applications. Security software, device-management tools, and required collaboration/sync applications should not be disabled casually. Optional applications can be evaluated if the user reports slow sign-in or excessive background activity.

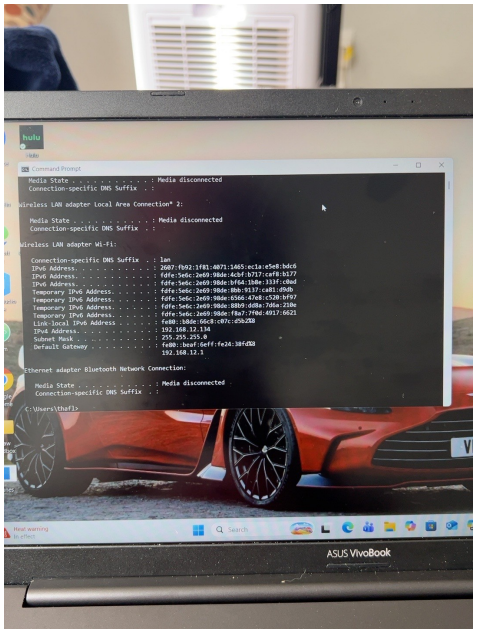
5. Network Troubleshooting - Commands and Results

A layered connectivity test was performed from Command Prompt. Each test isolates a different part of the network path, making it easier to identify whether a failure is local to the workstation, the local router, upstream Internet access, or DNS.

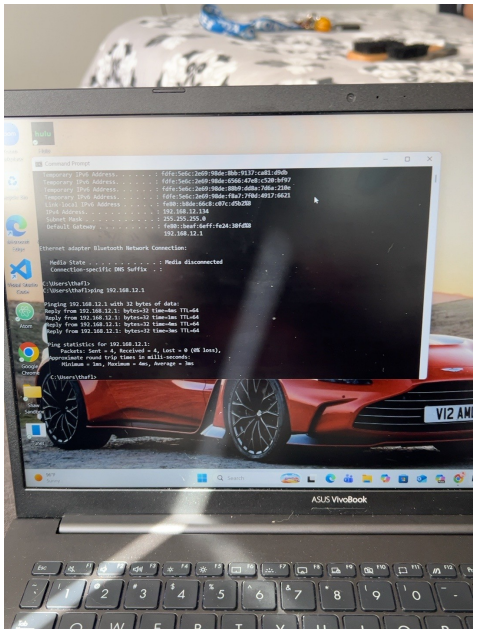
Command / test	Observed result	Interpretation
ipconfig	IPv4 192.168.12.134; subnet mask 255.255.255.0; default gateway 192.168.12.1	Valid private IPv4 configuration and gateway were present.
ping 192.168.12.1	4 sent, 4 received, 0% loss; average about 3 ms	Local default gateway was reachable.
ping 8.8.8.8	4 sent, 4 received, 0% loss; average about 27 ms	External IP connectivity was working.
ping google.com	4 sent, 4 received, 0% loss; average about 54 ms	Hostname resolved and external destination responded.
nslookup google.com	Returned multiple IPv4 and IPv6 addresses	DNS name resolution was functioning.

Conclusion: The captured results confirm a valid local configuration, successful communication with the gateway, successful upstream Internet reachability, and working DNS resolution.

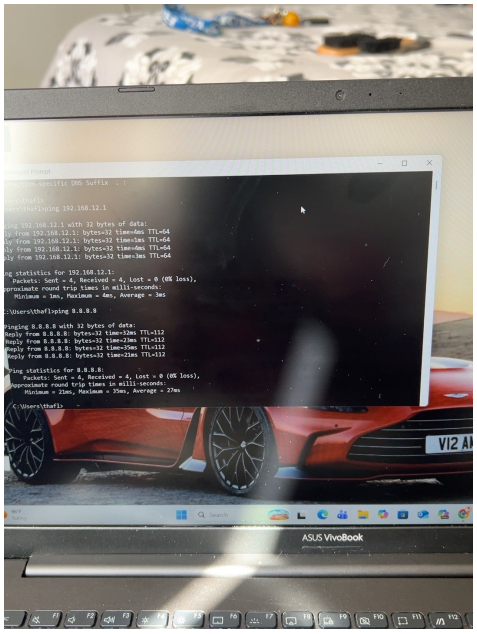
6. Network Evidence



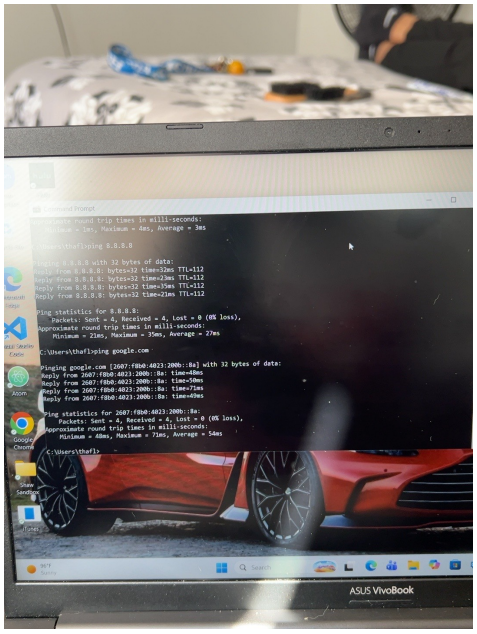
Evidence C - ipconfig output showing the Wi-Fi IPv4 address, subnet mask, and default gateway.



Evidence D - ping to default gateway completed with 0% packet loss.

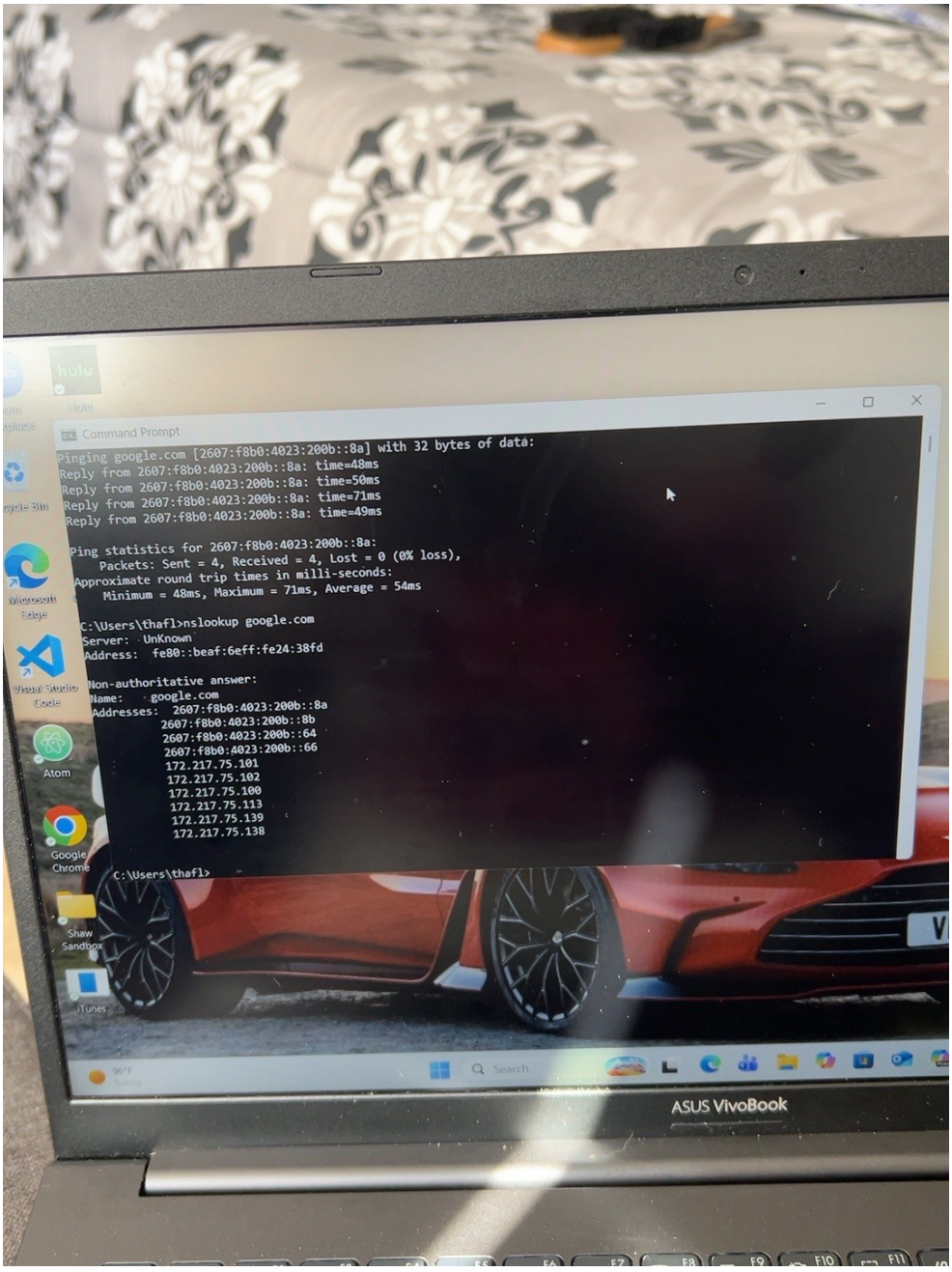


Evidence E - ping to 8.8.8.8 completed with 0% packet loss.



Evidence F - ping google.com resolved and completed with 0% packet loss.

7. DNS Verification Evidence



Evidence G - nslookup google.com returned multiple IPv4 and IPv6 records, confirming DNS name resolution.

The displayed DNS server name appears as "Unknown," but the lookup still completed successfully and returned valid address records. A missing reverse-resolvable server name does not by itself indicate that DNS resolution failed.

8. Service Desk Ticket Documentation

The following ticket-style entry summarizes the hands-on work completed in this lab.

Field	Documentation
Issue / request	Perform a Windows 11 workstation health and connectivity assessment for a service desk troubleshooting portfolio lab.
Initial checks	Collected system information and reviewed Task Manager performance and startup configuration.
Network actions	Recorded IP configuration; tested the default gateway; tested a known external IP; tested google.com; verified DNS with nslookup.
Findings	System resource usage was not saturated during capture. Valid IPv4 configuration was present. Gateway, Internet, and DNS tests succeeded with 0% packet loss on the documented pings.
Resolution / status	No active network fault was reproduced. Workstation connectivity and DNS resolution were verified as operational.
Verification	Successful ping responses from the local gateway, 8.8.8.8, and google.com; successful nslookup result; Task Manager and System Information evidence captured.
Escalation criteria	If a future fault persists, collect the exact error, check Windows Update/device status, review logs, and use elevated repair commands only when the symptoms justify them.

9. Completion Checklist

[X] Captured Windows system information and hardware baseline.

[X] Reviewed CPU, memory, disk, Wi-Fi, and GPU performance in Task Manager.

[X] Reviewed startup applications and their enabled/disabled status.

[X] Recorded local IPv4 configuration and default gateway.

[X] Verified communication with the local gateway.

[X] Verified external Internet connectivity using a known public IP.

[X] Verified hostname connectivity with google.com.

[X] Verified DNS resolution using nslookup.

[X] Documented findings in service desk ticket format.

[X] Included screenshot evidence supporting the completed scope.

FINAL LAB STATUS: COMPLETED

Scope note: This completed portfolio lab covers diagnostic assessment and network troubleshooting performed on October 6, 2026. DISM/SFC repair, Windows Update remediation, Device Manager remediation, and application/profile repair are retained as future expansion labs rather than being represented as completed actions.